Automatic Detection and Counting of Circular and Rectangular Steel Bars

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Abstract. The steel industry heavily relies on manual labor and the use of photoelectric sensors and complex counting machines to count steel bars. In the last decade, research on the automatic detection and counting of steel bars by using image processing and computer vision techniques have seen much progress. Nevertheless, most of past research focused mainly on circular shaped steel bars from a direct frontal camera angle. In this paper, we propose a method that is adaptable to both circular and rectangular shaped steel bars, and robust towards different camera angles and lighting intensity. The captured digital image first undergoes an essential pre-processing stage followed by edge detection which extracts the steel bar edges. For circular shaped steel bars, we apply Hough Transform followed by a post-process while the rectangular ones can be accurately found based on a series of morphological operations. Experiments conducted on a variety of challenging conditions demonstrate the capability of our approach to a good measure of success.

Keywords: steel bar detection, steel bar counting, circular steel bar, rectangular steel bar

1. Introduction

Steel manufacturing companies produce a wide range of steel products including billets, steel bars, carbon steel, wire rods, and steel pipes. Before these steel products are shipped out, they are normally counted. Popular methods include manual labor, photoelectric sensors and possibly, sophisticated and complex counting machines. These methods are tedious, costly, require a lot of human resources, and are highly prone to error. Despite a growing interest in vision-based automatic counting of steel bars in the last decade, most of past research focused mainly on circular shaped steel bars from a direct frontal camera angle and is less robust towards different angles, uncontrolled lighting setting and uneven stacking.

In this paper, we propose a practical and robust algorithm that can automatically count two types of the steel bars; circular and rectangular. The algorithm is performed in 3 steps; (1) pre-processing, (2) shape detection and (3) shape counting. In the pre-processing step, given an input image, we apply Gaussian blurring followed by morphological operations to distinguish the individual steel bars. Canny edge detector is then employed to produce the edge image. Next, in the shape detection step, the edge

map is processed using two variants of algorithms for circular and rectangular shaped steel bars respectively. This produces a set of connected components, each corresponding to the position of a single detected steel bar. In the shape counting step, the proposed algorithm finds the contours of the connected components and the number of distinct contours found is equivalent to final steel bar count in the image. Experiments were carried out on a variety of challenging conditions, and we were able to demonstrate the capability and robustness of our approach.

2. Related work

Generic approaches for detecting regular patterns in an image such as binary template matching [1] and lattices detection [2] are appealing solutions for the detection and counting of steel bars. However, an experiment conducted by Thammasorn et al. [3] demonstrated that direct application of these generic algorithms for steel bars counting can failed miserably. Thus, many researchers have proposed algorithms that are specific for steel bars detection and counting [3] – [8].

George and Wolfer [4] used a hybrid approach that combines ant colony optimization and particle swarm optimization to count small diameter tubular steel bars. This approach does not guarantee a solution when the stacked bars are not symmetrical, while memory consumption is a concern as it increases with the number of particles spawned. The work by Okumoto and Nakamura [5] reported an impressive result by estimating the circularity and cavity area, but their experiments appear to be inadequately conducted (merely 3 images used). Zhang *et al*. [6] proposed a mutative threshold segmentation approach where multiple thresholds are used to separate conglutinated sections. This method uses custom hardware, namely camera, light shade, reflector and light to produce good images. Round template matching is used to remove inter cover steel bars. Ying et al. [7] employed a series of image processing techniques such as chain coding, edge detection and improved Hough transform to automatically count steel bars. This approach succeeded in providing precise localization of steel bars but does not cater for non-circular shaped steel bars and does not tackle conglutination. Hou *et al*. [8] presented a multi-template covering algorithm based on connectivity and circularity to detect and count bundled steel bars. Their method achieved a high counting accuracy for images taken directly from a frontal angle. On a whole, these approaches focused mainly on circular shaped steel bars detection and are not robust towards different camera angles, and different diameter sizes in a single image.

More recently, a template based approach that can support both rectangular and circular steel bars was proposed [3]. This method produces good accuracy for four types of material objects; circular pipe, rectangular metal beam, square metal beam, and metal form under the couple of assumptions; camera angle must be direct, and the steel bar stacks should fill almost the entire image frame. Due to these image restrictions, this approach may not work for images taken from a slanted camera angle, bundles of circular bars that leave much empty space around them and bundles with uneven stacking.

3. Algorithm Description

In this paper, we focus on detection and counting of two types of steel bar; circular and rectangular. An overview of our proposed algorithm flow is shown in Figure 1. Generally, the algorithm is divided into 3 main steps: (1) pre-processing, (2) shape detection and (3) shape counting. The main difference between the respective algorithms for circular and rectangular steel bars lies in the shape detection step. The pre-processing and shape counting steps for both types of steel bar are similar. The following sub-sections describe these steps in detail.

3.1. Pre-processing

Given an input image of steel bars, the image is first converted to a grayscale

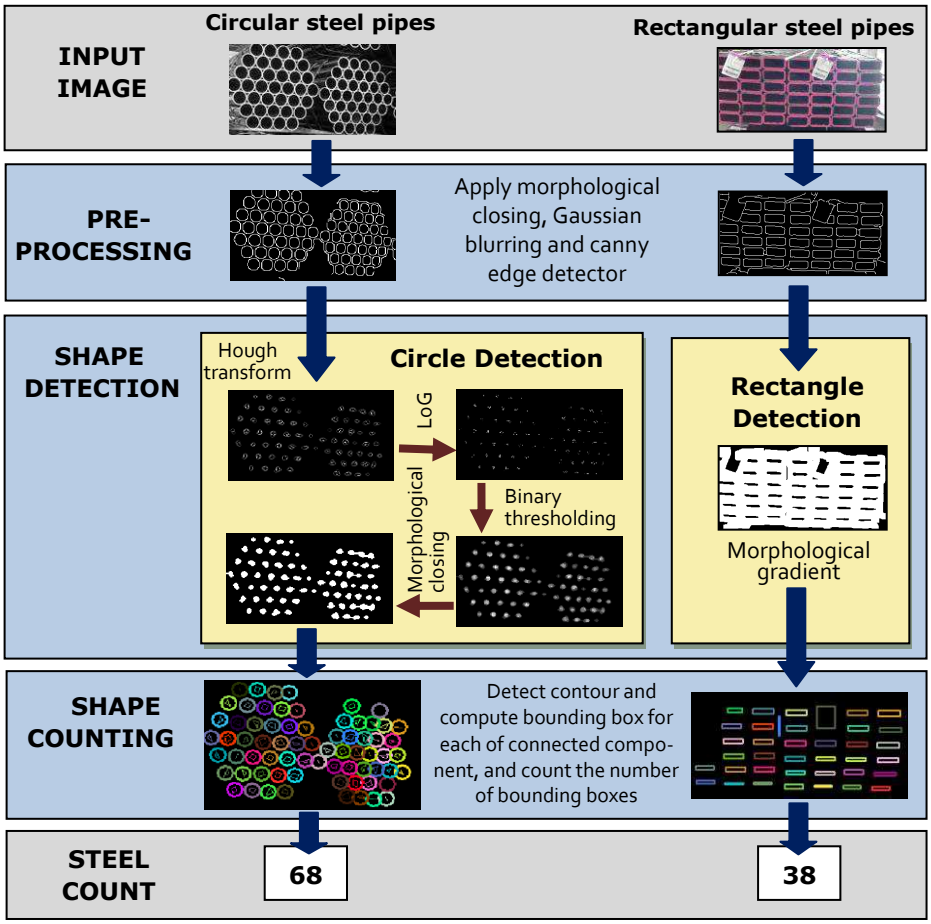

Figure 1: Overview of the proposed algorithm.

image. Then, Gaussian blurring is applied to reduce noise. In order to improve edge detection, morphological closing is used to separate steel bars from each other. This closing process has a positive side effect in which the out-of-focus steel bars and other unwanted objects are eliminated. The image is then passed to a Canny edge detector to extract the edges of the steel bars. The second block from the top in Figure 1 illustrates this pre-processing step.

The accuracy of the subsequent shape detection and counting algorithms depends significantly on the output of this pre-processing step. Therefore, to enhance the practicality of our approach, we allow users to adjust and pre-set the threshold values of the canny edge detector to obtain a good edge image. Besides, users can also provide the minimum and maximum radius of steel bars to be detected in order to further enhance the range of detection for the processed images.

3.2. Shape Detection

We proposed two variants of shape detection algorithms to detect both circular and rectangular steel bars. The output from this shape detection step is a set of connected components (regions or edge contours that are connected).

Circular shape detection. For circle shape detection, we first apply an improved Hough Transform [7] on the edge image. A Laplacian of Gaussian (LoG) technique is then applied to concentrate the weight of the interest points to the center of the steel bars. Next, we applied morphological closing to aggregate the interest points and create connected components that correspond to the position of each steel bar. Finally, we perform binary thresholding to obtain more refined and accurate connected components. The output of each individual process for the circular shape detection step is shown in Figure 1.

Rectangular shape detection. To detect rectangular shape, we perform two morphological operations, dilation and erosion on the edge image. The difference between these two resulting images is then computed to highlight the boundary edges of the steel bars. This morphological gradient process results in the formation of connected components that correspond to the position of the steel bars. Similarly, binary thresholding is then performed to obtain more refined connected components.

3.3. Shape Counting

Given the connected components from the shape detection step, our algorithm finds the contour of the connected components [9] and computes a bounding box/circle around each contour. As illustrated in the shape counting block in Figure 1, the location of the bounding boxes (polygons for the case of circular steel bars) indicates the position of each steel bar. Finally, our algorithm computes the number of bounding boxes/polygons to obtain the final steel bar count.

4. Results

Evaluation was done by testing on images provided by a steel manufacturing company. The steel bars in these images have been bundled up and are ready to be shipped out. Hence, a verification count is needed to validate the number of steel bars in each bundle. The test set consists of 10 images; 5 circular steel bar images and 5 rectangular steel pipe images, captured under different illumination settings and in a variety of challenging conditions, including slanting angles, inter cover, different diameter sizes in a single image and uneven stacking that causes the change of shape orientation. All the tested circular and rectangular steel bar images are shown in Figure 2 and Figure 3 respectively. Our approach takes from a few seconds to a maximum of 3 minutes (depending on image size) to count the number of steel bars in one image.

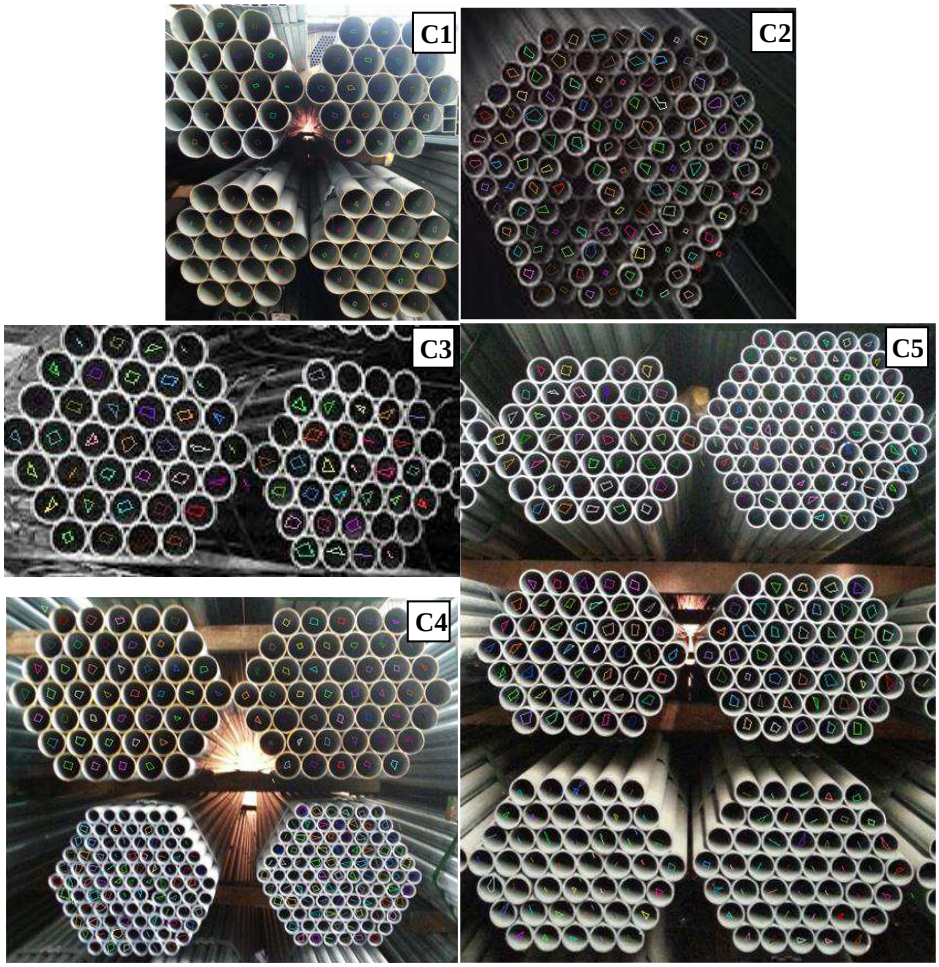

Figure 2. Circular steel bar image test set. Existence of colored polygon inside a steel bar indicates positive detection.

Figure 3. Rectangular steel bar image test set. Existence of colored square inside a steel bar indicates positive detection.

We use three standard metrics: Precision, Recall and F-measure, to evaluate the effectiveness of our approach. Precision is the fraction of retrieved steel bars that are correct. Recall is the number of retrieved steel bars divided by the number of actual steel bars. F-measure is the harmonic mean of precision and recall, which gives a good reflection of the overall accuracy of our algorithm.

Results showed that our algorithm is able to perform well under various conditions for both circular and rectangular steel bars. The average F-measure for the test set is 0.963, with precision slightly higher than recall. Comparatively, results of the image set for circular steel bars obtained higher average precision but lower average recall than the image set for rectangular steel pipes. Both circular and rectangular steel bars image sets achieve average F-measure value of more than 0.95, indicating that our proposed approach can be of good practical use.

The precision, recall and F-measure for each test image are reported in Table 1. For circular steel bar images, our algorithm performs very well for images captured from direct frontal view (**C1**, **C5**), regardless of diameter size. Both images achieved F-measure values above 0.97. Our test set also includes images captured under other challenging conditions as shown in Figure 2. In image **C2**, steel bars are not aligned properly, resulting in the *inter cover* effect [6], where illumination varies across steel

bars in a same bundle. Image **C3** is captured at a slanted camera angle, giving a perspective effect where steel bars in front appear to be bigger than steel bars at the back. Image **C4** and **C5** consists of bundles of steel bars with two significantly different sizes. Interestingly, compared to results of images with direct front view, the performance of these images (**C2**, **C3**, **C4**) are not far behind, all with F-measure scores > 0.94. Note that the precision for all three images is still high and the lower F-measure is due to lower recall. To further test our algorithm on its ability to tackle slanting angles, we artificially warp image **C3** to introduce a more extreme perspective angle before re-running our algorithm. There was a significant drop in performance, indicating that our algorithm can address slanting angles only to a certain degree.

Figure 3 shows the image set for rectangular steel bars. This set consists of images with either rectangular (**R1**) or square (**R2**, **R3**, **R4**, **R5**) shaped steel bars. The human process of stacking these rectangular/square shaped steel bars can sometimes result in seriously uneven stacking, as noticeable in images **R2** and **R3**. Unexpectedly, our

Table 1. Analysis of results on actual industrial steel bar images.

Image Label	Description of test image	Ground Truth	Steel Count	Precision	Recall	F-measure
Circular Steel Bar Images						
C1	Circular steel bars with big diameter	74	72	1.0	0.973	0.987
C2	Circular steel bars with slanted camera angle	74	68	0.985	0.906	0.944
C3	Circular steel bars with inter cover condition	127	115	0.991	0.898	0.942
C4	Circular steel bars with different diameter size	272	258	0.969	0.919	0.943
C5	Circle steel bars with small diameter	315	322	0.963	0.984	0.973
AVERAGE				**0.982**	**0.936**	**0.958**
Rectangular Steel Bar Images						
R1	Rectangular steel pipes with identification tags	38	38	0.948	0.947	0.947
R2	Rectangular steel pipes with uneven stacking front view	64	65	0.985	1.0	0.992
R3	Rectangular steel pipes with uneven stacking back view	64	63	0.968	0.953	0.961
R4	Rectangular steel pipes with uneven stacking	396	393	0.985	0.992	0.989
R5	Rectangular steel pipes with uneven lighting	433	405	0.983	0.919	0.950
AVERAGE				**0.974**	**0.962**	**0.968**
OVERALL AVERAGE				**0.978**	**0.949**	**0.963**

** **Image Label** indicates the reference to the corresponding images in Figure 2 & Figure 3.

algorithm performs exceptionally well for these images, scoring above 0.95 for precision, recall and F-measure. Performance for image **R1** is slightly lower due to the existence of tags blocking some of the steel bars. In our evaluation process, our ground truth dismisses the blocked steel bars but accuracy is still affected due to false positives generated by the tags. According to the steel company, the tags can be relocated such that no steel bars are blocked in order to improve its practicality for real-world implementation. Image **R5** presents an interesting test case where illumination varies across the image, with steel bars gradually become darker towards the bottom of the image. Our algorithm is able to obtain a good F-measure score of 0.95 for this image, an indicator of its robustness at handling illumination variations. Notably, steel bars that are significantly darker at the bottom of the image go undetected by our algorithm, as highlighted by the erroneous red box in image **R5** (see Figure 3).

Analysis of our results shows that our algorithm performs favorably well across a variety of challenging conditions. A good precision and reasonable recall level was reported for detection and counting of both circular and rectangular steel bars. The main contribution of our work lies in its adaptability to both circular and rectangular shaped steel bars and robustness towards considerable variations in camera angles, lighting intensity and steel bar diameter sizes. Besides, our algorithm also achieved remarkably good results for cases with uneven stacking.

5. Conclusion

In this paper, we propose a practical steel detection and counting algorithm that supports both circular and rectangular shaped steel bars. Experiments conducted on a variety of challenging conditions demonstrate the effectiveness and robustness of our approach. We believe that our algorithm is of good practical usage and can be employed as the cost-effective method to automatically detect and count steel bars in steel industries. In the future, we intend to further enhance the robustness of our algorithm towards extremely slanted camera angles and a large variety of lighting conditions. Besides, we would also extend our approach to other types of steel bars such as rods and billets, while working towards improving the overall speed of our algorithm.

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